IM3180 Design and Innovation Project (AY2025/26 Semester 1)
Individual Report

Name: Poluri Sai Sriya

Group: IE08

Project Title: DeciWise

Contributions to the Project and Report (1-2 pages)

Project

- Initial Flowchart development for a better understanding of our project
  - Designed the initial multilayer branching for decision making.

- Wireframing in Figma and helping out with the UI Design

- Research on AI Models to be used and 3D Image generation

- Scenario Flowchart development (creation of JSON File to interconnect nodes)
  - Defined the core logic, node types and tier structure needed for consistent node generation
  - Developed the IDs and linking format in JSON file.

- Development of Flask API for integration to the frontend
  - Built the whole AI flask app, which created separate routes for flowchart generation, image generation and AI suggestions

- OpenAI API connection development
  - Prompt Engineered the scenario, option and ending generation to produce a smooth flowchart into a JSON format, enabling a flattened storyline without the need for visual diagrams.
  - Designed how AI generated text should map to nodes.
  - Design the flowchart based on the tier levels selected by the user.

- Psychological Dimensions Integration
  - Prompted GPT unique psychological dimensions assignments for each node in the scenario and options text in the flowchart.
  - Ensured consistent tone and behaviour by prompting GPT.

- AI Image Generation System
  - Integrated the Gemini Model for Image generation
  - Prompted the style and theme of image generation to avoid inappropriate images so that it can stick to the theme.
  - Encoded and Decoded to Base64 for smooth transfer to frontend without corrupting the image.

- AI Suggestions Development
  - Implemented AI-driven suggestion engine that helps trainers generate new branching points for a scenario that users can use.

- Integration of code with Frontend
  - Ensured and adjusted the code as needed so that the backend output can match the UI's expected data format correctly.

<u>Report (indicate pages and appendixes written by you for the Group Report)</u>

- Section 3.5: Overview of Flask (backend) and React (frontend) frameworks
- Section 4.1: Design Consideration / Choice of Components / Technology Stack (Section 4.1.2: Backend)
- Section 4.1: Design Consideration / Choice of Components / Technology Stack (Section 4.1.4: AI Model: GPT-4 / GPT-5/ Nano-banana (for story content and node suggestions))
- Section 4.3.2: AI Generation Pipeline
- Section 4.4.2: API testing (Flask Endpoint Validation)
- Section 5.1. Maintaining JSON integrity from AI output
- Section 5.2. AI not understanding prompts
- Section 5.3. Generating Images along with encoding and decoding to Base64
- Section 5.7. Integration of AI Flask APP with Frontend

<u>Other Deliverables like Project Chater (indicate pages and appendixes), Poster, Presentation</u>

- Project Charter:
  - Project Purpose or Justification
  - Project Management Plan

<u>Slides (indicate slide numbers) and Video, if applicable</u>

Reflection on Learning Outcome Attainment

Reflect on your experience during your project and the achievements you have relating to at least two of the points below:

(a) Engineering knowledge
(b) Problem Analysis
(c) Investigation
(d) Design/development of Solutions
(e) Modern Tool Usage
(f) The Engineer and the World
(g) Ethics
(h) Individual and Collaborative Team Work
(i) Communication
(j) Project Management and Finance
(k) Lifelong Learning

Point 1: Modern Tool Usage

Throughout this project, I gained extensive experience using modern tools that are widely adopted in the industry. Working with Flask, React, OpenAI and Gemini required me to understand how these frameworks interact and how AI models can be integrated into real world applications. I learned to build various APIs, routes, validate endpoints using Postman and work with AI models for both text and image generation.

Developing the flowchart text, prompting the AI to give it according to the necessary outputs needed was when I had realised I had maximised my coding skills. To prompt the AI, you had to be specific with each and every line that you are prompting it otherwise there would definitely be a problem once generated. I had also realised that AI cannot do everything resulting in me hardcoding a few parts such as the JSON formatting, node IDs and text formatting for a better user experience.

Moreover, generating images especially into Base64 Strings without corrupting the image was a new concept for me. Generating images separately felt easier however, generating images within the Flask App allowed me to learn new skills, making the debugging process my learning tool rather. Since I was responsible for the whole Flask App, I had to make sure I maximised the Flask routes and the Postman endpoints as I did not have the frontend to test how the AI worked. It was quite a new experience to learn all the Postman endpoints required for all the testing I needed to perform.

I also learned how to use Git and Github from this project where I had to keep committing my changes to my branch. I had no proper experience of using Github before so it was quite challenging especially when there were many errors when merging the branches. Initially, each person had a separate branch to themselves which was quite easy to understand. However, when integrating the code, that was when unexpected problems arose.

Overall, the project strengthened my ability with coding logic and my ability to use modern software engineering tools in a practical way possible. More importantly, it taught me how to integrate different technologies, i.e AI models, backend services, etc. into a cohesive system.

Point 2: Individual and Collaborative Teamwork

This project required close collaboration and clear communication with one another. While each member had specific responsibilities, our work was deeply interconnected. Unlike IDP which was a pair work, DIP required constant communication of updates otherwise it would have led to a collapse.

We had weekly discussions and meetings, always brainstorming on what we could do more. I personally enjoyed our discussions because this was the one of the few times where I focused those 3 hours without any distractions. Through these meetings, I have also learned many new skills and new technical terms and easy logical ideas to code properly.

Initially we planned to work on our parts separately, however, once it came to integrating everything, we realised we should have approached this problem differently. I learned how important it is to communicate even more early, test frequently and be flexible when teammates need adjustments to support their part of the system. Although we had 13 weeks and communicated at least two or three times in a week (which felt sufficient), we still faced a few hiccups along the way. In the end, after a few nights of spending in each other's halls, spending some time in SS benches, the admin building meeting room, we managed to push through and make it work together.

Overall, compared to the other times I have worked in a group project, I am genuinely glad to have been paired up with this team. Everyone consistently pulled their own weight, stayed accountable and completed their tasks on time. This project not only gave me a fun learning experience but also strengthened my teamwork skills, especially in coordinating technical decisions and supporting each other through challenges.

Acknowledging/Declaring the Use of GAI

Please refer to NTU's Current Policy & Guidelines on the Use of Generative AI available in NTUlearn home page and the link: https://entuedu.sharepoint.com/sites/Student/dept/ctlp/SitePages/Exploring-the-Impact-of-Generative-Artificial-Intelligence-(GAI)-Tools-on-Education.aspx

- Complete the following declaration if applicable.
- Create a Paper Trail to document the input prompt, output obtained, and how you have used it

I Poluri Sai Sriya (student name), SAISRIYA001@e.ntu.edu.sg (NTU email) honestly and sincerely make the following declaration in relation to the following course submission:

1. Name of course: Design & Innovation Project
2. Course Code: IM3180
3. Instructor: Prof Andy Khong
4. Title of Assignment/Project Submission: DIP Individual Report

In relation to the foregoing I hereby declare that, fully and properly in accordance with the Assignment/Project Instructions I have (check where appropriate):

i. Used GAI as permitted to assist in generating key ideas only. ☐
ii. Used GAI as permitted to assist in generating a first text only. ☐
And/or
iii. Used GAI to refine syntax and grammar for correct language submission only. ☐
Or
iv. As it is not permitted: Not used GAI assistance in any way in the development or generation of this assignment or project. ☐

I also declare that I have :

a. Fully and honestly submitted the digital paper trail required under the assignment/project instructions; and that
b. Wherever GAI assistance has been employed in the submission in word or paraphrase or inclusion of a significant idea or fact suggested by the GAI assistant, I have acknowledged this by a footnote; and that,
c. Apart from the foregoing notices, the submission is wholly my own work.

Student Name & Signature

14th November 2025
Date